

QUESTION BANK

CHAPTER 1 INTRODUCTORY CONCEPTS

PAGE No: 11

- 1) What is analog and digital system? (2016,2008)
- 2) Write down the advantages of digital techniques. (2016)
- Or, what are the advantages of digital techniques over analog? (2008)
- 3) Concisely describe the major difference between analog and digital quantities. (2010)
- 4) What steps should be followed in digital techniques while dealing with analog world? Discuss with an example. (2009) (2010)
- Or, Explain with diagram how digital world interface with analog world. (2009) (2010)
- 5) Describe the main functional units of a digital computer. (2015, 2011) (2008)
- Or, Describe the organization of a digital computer with figure. (2015, 2011) (2008)
- 6) Describe the relative advantages of parallel and series transmission of digital data. (2012)
- 7) What is binary number system? How any binary number can be converted into its equivalent decimal number and vice versa? (2017,2013)
- 8) What are the advantages of digital technique over analog? Or, what are the advantages of digital technique. (2016,2008,2018)
- 9) Define digital system. Write down importance of digital system. [2019]
- 10) Discuss the major parts of a digital computer with required diagram. [2019]

CHAPTER 2

NUMBER SYSTEMS AND CODES

PAGE No: 19

- 1) Convert the following numbers into given base numbers: (2015)
 - I. $(562.65)_{10} = (?)_2 = (?)_8$
 - II. $(BCD)_{16} = (?)_8$
- 2) Convert the following numbers into given base numbers: (2017)
 - i. $(110101)_2 = (?)_{10}$
 - ii. $(427.24)_8 = (?)_2$
 - iii. $(129A.B8C)_{16} = (?)_2$
 - iv. $(225.225)_{10} = (?)_2$
 - v. $(11010111.110)_2 = (?)_{16}$
- 3) Convert the following numbers into given base numbers:- (2013)
 - a) $(1001001.011)_2 = (?)_{10}$
 - b) $(225.225)_{10} = (?)_2$
 - c) $(623.77)_8 = (?)_{10}$
 - d) $(11010111.110)_2 = (?)_{16}$
 - e) $(2AC5.D)_{16} = (?)_8$
- 4) Convert the following operation:- (2008)
 $(526.56)_{10} = (?)_2 = (?)_8 = (?)_{16}$
- 5) Represent the decimal number 396 in binary form (2011)
 - i. Binary Code.

- ii. BCD Code
- iii. Excess-3 Code.

6) Represent the decimal number 17 in binary form using

- I. BCD
- II. Excess-3
- III. Octal code

7) How many bytes are needed to represent 23510 in binary? (2016)

8) Add and multiply the numbers 11110 and 111. (2016)

9) Discuss the repeated division method for converting decimal number to binary number system of integer with a flow-chart. (2015)

10) What is the largest decimal value that can be represented in BCD using two bytes? (2016)

11) How any binary number can be converted into its equivalent decimal number and vice versa?

12) Convert decimal 378 to a 16 bit binary number by first converting to hexadecimal number system. (2016)

13) What is a code converter? Design a 4-bit Binary-to-gray code converter. (2012)

14) What is binary number? Briefly explain. (2013)

15) What is binary code? Why is it necessary? (2017)

16) Define the following terms with example:

- 1. ASCII code (2019,2015,2012,2009)
- 2. Excess 3 code (2015)
- 3. Gray code [2019](2012)
- 4. Hexadecimal code; (2009)
- 5. Byte (2008)
- 6. Octal. (2009)
- 7. BCD (2008)

17) What is parity bit? (2016, 2008)

18) How many types of parity bits are there? Write down a truth table for even parity bit generator for all possible 4-bit input. (2009)

19) Why parity bit is significant for transmission of digital signal? (2011)

20) Why can not the parity method detect a double error in transmitted data? (2010)

21) Difference between Binary and BCD code. (2016)

22) Distinguish between Binary and decimal number system; (2014)

23) Distinguish between Octal and hexadecimal system; (2014)

24) BCD Vs Gray code

25) Explain the method of subtraction using 2's complement method with example (2017)

26) Using 2's compliment systems subtract 5 forms 9. (2016)

27) Perform the following operation using 2's compliment method: (2012)

i) $48 - 23$ ii) $48 - (-23)$

28) What is Unicode? (2020)

29) Draw and explain BCD to 7 segment decoders. [2019]

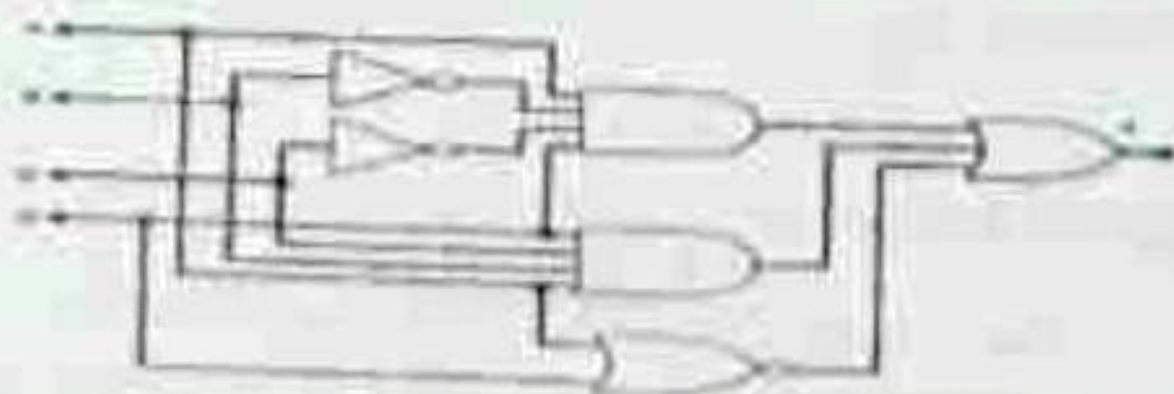
LOGIC GATES AND BOOLEAN ALGEBRA

- 1) Write the name of digital logic gates. Briefly explain the diode logic gate. (2016)
- 2) Explain XOR gate and XNOR gate with their logic symbol, Boolean expression and truth tables (2014, 2011)
- Or, Write down the logic symbol, Boolean expression and truth tables for the following logic gates:

- a) AND
- b) OR
- c) X-OR
- d) X-NOR

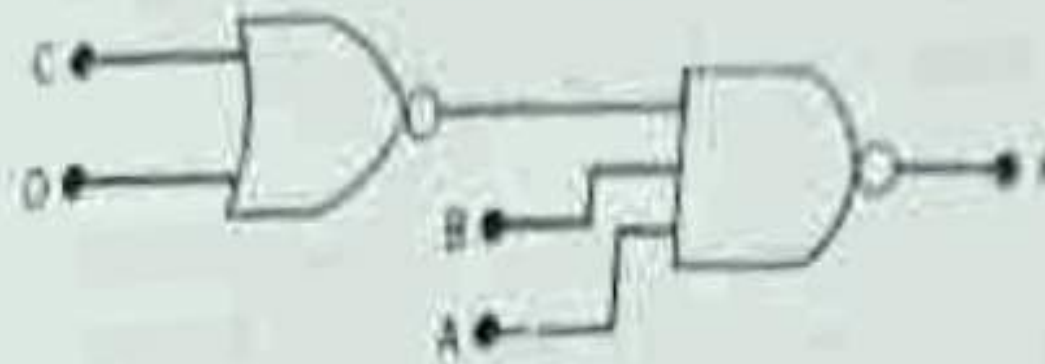
- 3) What is universal gate? (2019, 2013, 2017, 2014)
- 4) Prove the universality of NAND gate and NOR gate. Or, Show that the basic gates can be realized by the NAND and NOR gates. (2019, 2017, 2014, 2013, 2010, 2008)
- Or, Why NAND gate is called universal logic gate? Prove the universality of NAND gate. (2012)
- Or, Discuss the universal property of NAND and NOR gates. (2010)
- 5) Show that how a two input NAND gate can be constructed from two input NOR gate.
- 6) Implement AND OR logic gate by using NOR gate only. (2008)
- 7) Implement AND OR logic gate by using switches only. (2009)
- 8) Define truth table. (2013)
- 9) Draw truth table and the AND OR logic circuit for the Boolean expression $Y = \bar{A}B\bar{C} + \bar{B}C$
- 10) Explain Boolean Postulates.
- 11) Use De-Morgan's theorem convert the Boolean expression to its min term form, $(A + \bar{B} + C + D)(A + \bar{B} + \bar{C} + D) = Y$ (2011, 2009)
- 12) Using Boolean expression show that $F \cdot \bar{F} = 0$ if $F = \bar{x}y + xy\bar{z}$ (2008)
- 13) Using Boolean expression prove that $x + y = x + \bar{x}y$ (2016)
- 14) Implement the logic circuit that has the expression $x = \overline{AB(C + D)}$ using only NOR and NAND gates. (2013)
- 15) Determine the output expression for the following circuit and simplify it using De-Morgan's theorem. (2017)
- 16) Draw the three graphic symbols for inverter.
- 17) Define inverter.
- 18) What is Boolean algebra? Explain the Boolean function $F = A + BC$ in a sum of min terms. (2017)
- 19) What is Boolean algebra? Explain the Boolean function $F = A + \bar{B}C$ in a sum of min terms. (2020)
- 20) Write down Demorgan's theorem. Prove Demogon's theorem with 3 variables. (2019)
- 21) Write down the truth table for a full adder. Draw the logic circuit and explain its operation. (2019)

- (1) What do you mean by sequential circuit? Explain with necessary diagram. [2014]
Or, what do you mean by sequential logic circuit and combinational logic circuit? [2010]
- (2) What are the difference between combinational logic circuits and sequential logic circuit? [2019][2008]
Or, what are the differences between combinational and sequential circuits? [2014]
- (3) Design a logic circuit that output is high only when a majority of inputs A, B, and C are low.
- (4) Design a logic circuit with inputs P, Q, R so that output S is HIGH whenever P is 0
Or whenever $Q=R=1$ [2012]
- (5) Design a logic circuit that has three inputs A, B and C And whose output will be high when a majority of the inputs is high. [2010,2015]
- (6) Implement a logic circuit having output expression $z = (\overline{A + B + C}) \cdot \overline{B}C$ [2011]
- (7) What is K-map? Explain it with suitable example [2016,2015]
- (8) Mention the limitation of K-map method. How can this limitation be overcome? [2017]
- (9) Minimize the following function using K-map- [2015]
 $f(A, B, C, D) = ABC\overline{D} + \overline{A}BCD + \overline{A}\overline{B}C + \overline{A}BD + A\overline{C} + A\overline{B}C + \overline{B}$
- (10) Using K-map simplify the following expression:- [2013]
 $Y = \overline{A}\overline{B}\overline{C} + \overline{B}C + \overline{A}B$
- (11) Minimize the following expression using K-maps [2011]
 $f(A, B, C, D) = \sum m(0,1,2,3,5,7,8,9,11,14).$
- (12) Minimize the following function using K-map and realize using NAND only: $F(A, B, C, D) = \sum(1,3,5,8,7,11,15) + D(10,13)$ [2018,2010]
- (13) What do you mean by 'don't' care condition? [2017]
Or, States don't care condition. [2014]
- (14) Simplify the Boolean function, $F(A, B, C, D) = \sum(1,3,6,7,11,15)$ and don't care conditions, $d(A, B, C, D) = \sum(0,2,4,5).$ [2017,2016,15]
- (15) Minimize the following expression using K-maps
 $f(A, B, C, D) = \sum m(4,5,6,7,8,12) + d(1,2,3,9,11,14)$ [2014]
- (16) Draw a circuit using the following expression: [2013]
 $X = \overline{A}B + A\overline{B}$
- (17) Determine the output expression for the following circuit and reconstruct the circuit of its simplified expression:-

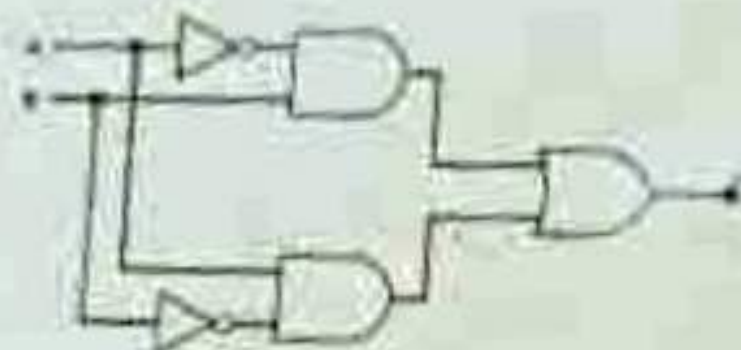


- (18) Simplify the following Boolean expression:- $Y = \overline{P}R(\overline{P}QS) + \overline{P}Q\overline{R}S + P\overline{Q}R$ and implement it with basic logic gates. [2015]
OR, $Z = \overline{A}C(\overline{A}BD) + \overline{A}B\overline{C}D + A\overline{B}C.$ [2012]

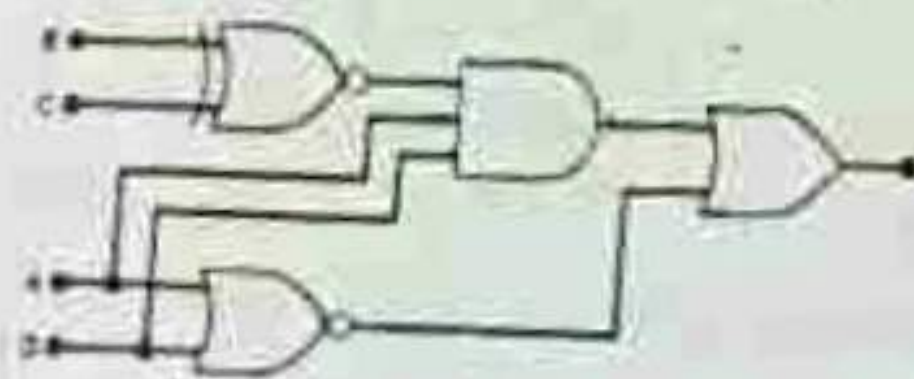
- (19) What is absorption law? Explain them. (2017)
 (20) What is venn diagram? Explain it for two variables. (2016)
 (21) What is Boolean algebra? Explain the Boolean function $F = A + BC$ in a sum of min terms. (2017)
 (22) State and prove involution law. (2017)
 (23) Prove that $x + x = x$ and $x \cdot x = x$. (2016)
 (24) Define min term and max term. (2008,2011,2013,2016)
 (25) State and prove De-Moegan's theorem. (2017)
 Or, State and prove De-Moegan's theorem and prove them given truth table for three variables.
 (26) Determine the output level in the following figure for $A=B=C=1$ and $D=0$. (2017)



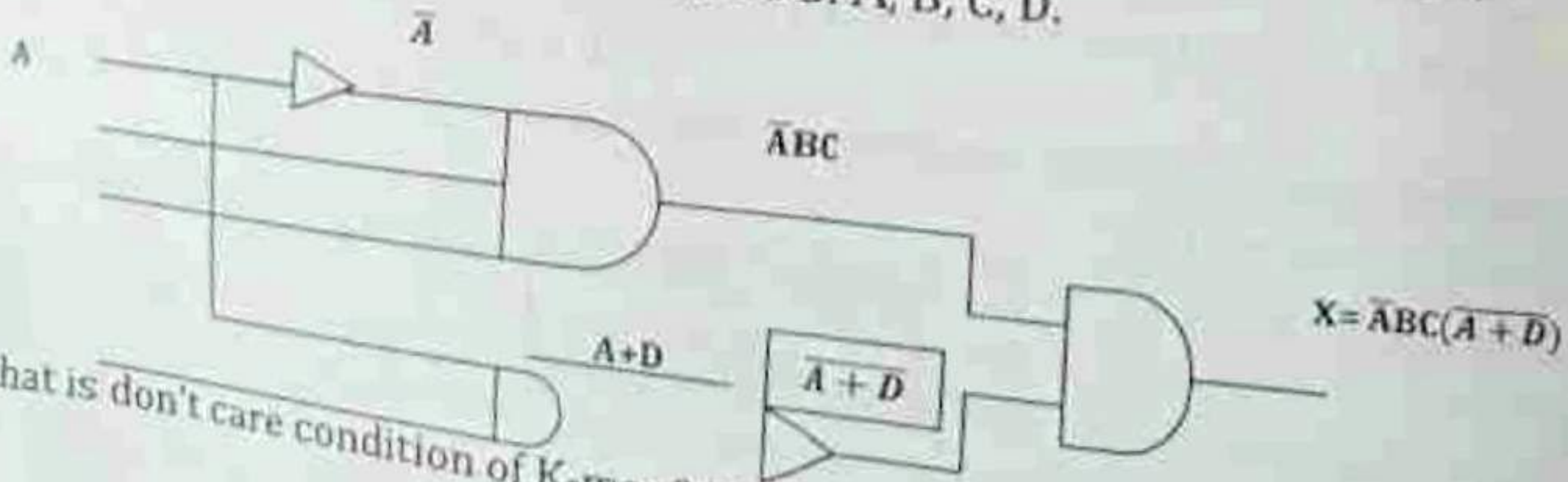
- (27) Determine the output expression for the following circuit:- (2011)



- (28) Determine the output expression for the following circuit:- (2020,2012)



- 29) What is karnaugh map? Using k-map simplify the following expression:-
 $F(A, B, C, D) = \sum m(1,3,4,6,8,9,11,13,15) + d(0,2,14)$ [2020]
 30) Define K-map. Using K-map simplifying the expression, $y = \bar{C}(\bar{A}\bar{B}\bar{D} + D) + A\bar{B}C + \bar{D}$,
 also draw the logic diagram of simplified y. [2019]
 31) Consider the following Boolean logic diagram. If you want to get the output value
 $x=0$ and $x=1$ then what will be the input values of A, B, C, D.



- 32) What is don't care condition of K-map? Explain. (2019)

CHAPTER 5

FLIP-FLOPS AND RELATED DEVICES

PAGE No: 71

1) Define the following terms:

- ✓ Asynchronous system
- ✓ Synchronous System
- ✓ Clock Signal
- ✓ Set up time and Hold time
- ✓ Flip-flop and Latch

[2015,2016]

[2016,2012,2011]

[2013]

2) Write down the applications of flip-flop.

3) Define NOR gate latch. Explain NOR gate latch with truth table and diagram.

[2014, 2013, 2009]

4) Define NAND latch

[2009]

5) Explain with proper diagram of a clocked S-R/S-C flip-flop.

[2014,2013]

Or, Explain the logic operation of the basic S-C latch

[2011]

6) Draw the logic circuit of an S-R flip-flop with NAND latch and explain its logic operation.

[2012][2019]

Or, Draw S-R FF using NAND gate and explain its operation.

[2010]

7) Discuss the internal circuit of the edge triggered J-K flip-flop with diagram.

[2010]

Or, Draw the internal circuit diagram of a JK flip-flop and explain its operations.

[2016]

Or, Describe the JK flip-flop with internal diagram.

[2008]

8) How can a JK flip-flop be modified to operate as a D flip-flop?

[2016]

9) How an S-R flip-flop can be converted into J-K flip-flop? Discuss with ckt-diagram.

[2014]

10) Briefly explain the clocked S-C flip-flop with suitable diagram.

[2016]

11) How can a J-K flip-flop be modified to operate as a D flip-flop and a T flip-flop.

[2020]

CHAPTER 6

PAGE No: 79

DIGITAL ARITHMETIC: OPERATIONS AND CIRCUITS

1) Describe ALU with diagram.

[2008]

2) Describe half adder circuit with diagram.

[2013]

3) How two half adder circuits can be used as a full adder?

[2013]

Implement a full adder circuit.

[2008]

Or, Write down the truth table for a full adder. Draw the logic circuit and explain its operation.

[2010]

Or, Design a full adder circuit and explain its operation with truth-table.

[2012]

Or, Design and construct a full adder circuit.

[2013]

4) Design and explain a 4-bit parallel adder circuit with registers.

[2017]

5) Distinguish between half subtractor and full subtractor.

[2011]

Or, Define half subtractor and full subtractor

CHAPTER 7 COUNTERS AND REGISTERS

PAGE No: B5

- 1) What is counter? (2013, 2011)
- 2) Explain a 4-bit asynchronous counter with figure. (2013)
Or, Describe the asynchronous counter with its circuit's operation. (2016)
- 3) Differentiate between synchronous and asynchronous counter. (2015, 2014, 2013, 2011, 2009)
Or, what are the advantages and disadvantages of a synchronous counter over an asynchronous counter? (2012)
- 4) What is the difference between the counting sequence of an up counter and a down counter? (2010)
- 5) What is shift register? (2015, 2014, 2013, 2011, 2008)
- 6) Explain 4-bit shift register with wave form. (2015, 2008)
- 7) Draw and explain the logic operation of 3-bit serial-in, serial-out shift register with timing diagram. (2013, 2014, 2011)
- 8) Explain MOD-8 down counter with suitable figure. (2017)
- 9) Describe with diagram the principle operation of a MOD-10 counter. (2008)
Or, Describe MOD-10 counter with transitional diagram (2015)
- 10) Design a MOD-10 ripple counter from a MOD-16 ripple counter by using J-K FF and show the waveforms and truth table. (2009)
- 11) Write down the operation of a 4-bit Johnson counter with necessary diagram. (2015)
- 12) What is transitional diagram? (2015)
- 13) Define setup time, hold time, propagation delay and MOD-10 number? (2015)
- 14) Draw the logic diagram of a 4-bit ring counter and explain its operation (2011)
- 15) Describe the operation of a 3-bit asynchronous binary counter. (2011)
- 16) Draw and explain serial data transfer from register to register. (2016)
- 17) Show how to wire a 74LS 293 as a MOD-14 counter. (2010)
- 18) What do you mean by propagation delay of Ripple counter? Explain in brief. (2017)
- 19) Design and discuss the MOD-6 counter. [2019]

CHAPTER 8 MSI LOGIC CIRCUITS

PAGE No: 102

- 1) What is Decoder?
- 2) Mention some applications of decoders. (2011, 2012, 2013, 2016)
Or, Mention the use of Decoder. (2012)
- 3) Realize a 4×16 decoder by using two 3×8 decoders. (2011)
- 4) Realize a 4×16 decoder by using two 3×8 decoders. (2014)
- 5) Explain a 3 to 8 line Decoder with truth table and circuit diagram. (2014)
- 6) Explain BCD to 7 segment decoder. (2016, 2013, 2011, 2009, 2008)
- 7) Explain a BCD decoder with truth table and circuit diagram. (2016)
- 8) What is encoder? (2015, 2009)
- 9) Mention the use of ENCODER. (2012, 2014, 2015, 2009)

10) Write the advantages of a priority encoder

11) What are the Drawbacks of Encoder

(2015, 2009)

12) Explain an octal to binary encoder with truth table and circuit diagram.

13) Explain 4 to 2 Encoder. Explain Priority Encoder

(2012, 2014)

14) Design a decimal-to-BCD priority encoder.

15) What is multiplexer?

(2010)

16) Design 8:1 multiplexer with truth table

(2011, 2008)

17) Draw a logic circuit of a basic two input multiplexer.

(2008)

18) How 16:1 Multiplexer can implement from 8:1 Multiplexer?

(2017)

Explain with diagram.

(2009)

19) What are the applications of Multiplexer and De-multiplexer?

(2008)

20) Design and explain four input multiplexer.

(2020)

21) What are the difference between multiplexer and de-multiplexer.

(2020)

22) Define MUX and DMUX. Why MUX is called data selector? and explain it.

(2019)

CHAPTER 9

INTERFACING WITH THE ANALOG WORLD

PAGE No: 118

1) What is transducer?

(2017)

2) State the purpose of analog-to-digital conversion.

(2012)

3) Describe the operation of an analog to digital converter.

(2008)

Or, Discuss the working principle of an A/D converter with required diagram.

(2016)

4) Discuss the working principle of D/A converter with required diagram

(2015)

Or, Describe the operation of a D/A converter with circuit diagram.

(2013, 2009)

Or, Explain with a diagram of a D/A converter circuitry.

(2017)

Or, Discuss the basic operation of DAC circuit with proper CKT.

(2014)

5) Write down the characteristics of a DAC.

(2014)

6) Differentiate between A/D and D/A conversion.

(2017)

7) A five bit DAC has a current output. For a digital input of 10100, an output current of 10 mA is produced. What will I out be for a digital input of 11101?

(2017)

8) A 10-bit DAC has a step-size of 10mv. Determine the full-scale output voltage and the percentage resolution.

(2010)

9) What is the largest value of output voltages from an eight-bit DAC that produces 1.0 v for a digital input of 00110010?

(2019)(2016)

10) Explain the operation of a 4-bit R-2R ladder DAC

(2012)

Or, Briefly explain a basic R/2R ladder DAC.

(2017)

11) Define SAC. What are the main advantages of an SAC over a digital ramp converter?

(2015)

12) What is quantization?

(2014)

3) What is step size and resolution?

(2009)

4) Explain the operation of successive approximation ADC with required diagram.

(2019)

5) Mention the advantageous of successive approximation ADC 3 over digital Ramp ADC.

(2019)

Tanjuu

- 16) What is encoder? Explain the working principle of 8 line to 3-line encoder with logic diagram and truth table. [2019]

CHAPTER 10

MEMORY DEVICE

PAGE No: 132

- 1) Describe semiconductor memory technologies
- 2) Describe different types of RAM
Or, Describe different types of DRAM
Or, Describe static and dynamic RAM
Or, Describe SRAM and DRAM
- 3) Write down the application of RAM and ROM
- 4) A certain semiconductor memory chip is specified as $2K \times 8$. How many words can be stored on this chip? What is the word size? How many total bits can this chip store?
- 5) Which memory stores the most bits: a $5M \times 8$ memory or a memory that stores 1M words at a word size of 16 bits?
- 6) Explain the general memory operation
- 7) Describe the conditions at each input and output when the contents of address location 00100 are to be read.
- 8) A certain memory has a capacity of $4K \times 8$.
- 9) How many data input and data output lines does it have?
- 10) How many address lines does it have?
- 11) What is its capacity in bytes?
- 12) Draw $16K \times 8$ ROM.
- 13) What is semiconductor memory? Discuss the classification of semi conductor memory.
- 14) Define memory. Discuss the basic organization of a memory unit of 3×4 bits. [2020]
- 15) Differentiate between EPROM and DRAM. [2019]
- 16) How many $32k \times 8$ RAM chips are needed to provide a memory capacity of 256k bytes? [2019]

CHAPTER 11

PAGE No: 141

INTRODUCTION TO VHDL AND CAD TOOLS

- 1) What do you mean by VHDL?
- 2) What is VHDL module
- 3) Why VHDL is used?
- 4) Describe a design of VHDL.